Introducing Angiogenesis in Predictive Models of Cancer Development

Student: Hossam El-Basiouny

Supervised by Matilde Palmeira, Rui Travasso

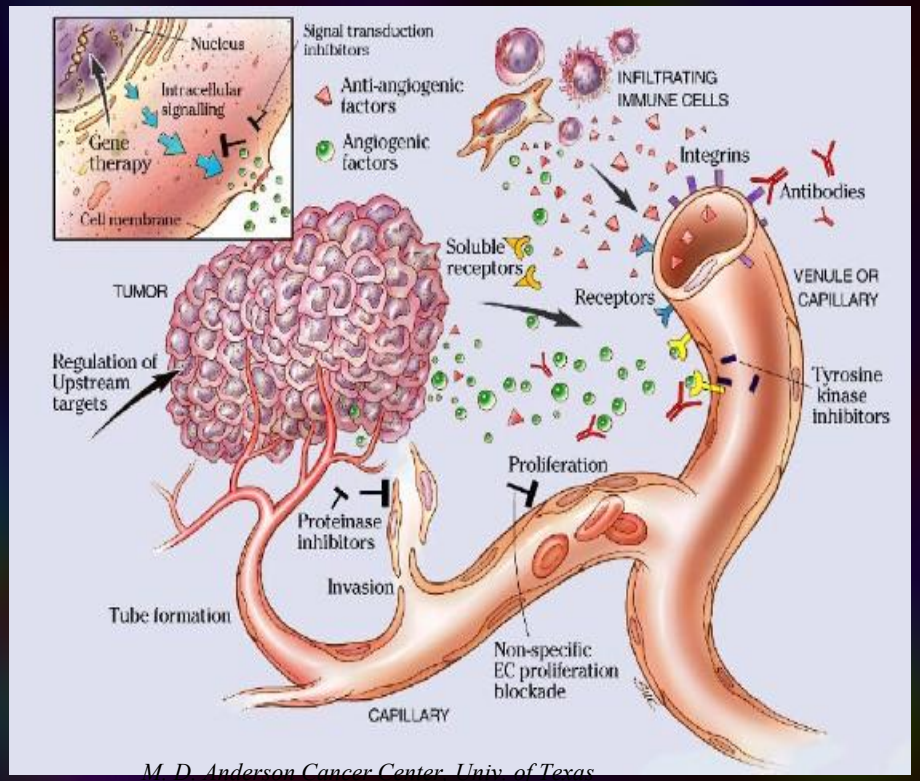
Tumors & Angiogenesis

Angiogenesis is the process by which new blood vessels from existing ones

Angiogenesis is strongly correlated with solid tumor malignity and is driven by tumor hypoxia

Relevant in varied situations in health and disease

- Morphogenesis
- Inflammation
- Wound healing
- Diabetic Retinopathy
- Over 50 other pathologies



M. D. Anderson Cancer Center, Univ. of Texas

- The goal is to explore simple predictive models of tumor growth in 2D and that includes how irrigation affects its development

- The model results show how the cells move, how they grow and can be fit from longitudinal medical imaging

Tumor Cell Density

$$\partial_t n = D\nabla^2 n + (\alpha_0 + \alpha_1(v^\alpha - 1)) \cdot n \cdot \left(1 - \frac{n}{N_{max}}\right)$$

Diffusion

Irrigation

Proliferation

Vessel Density

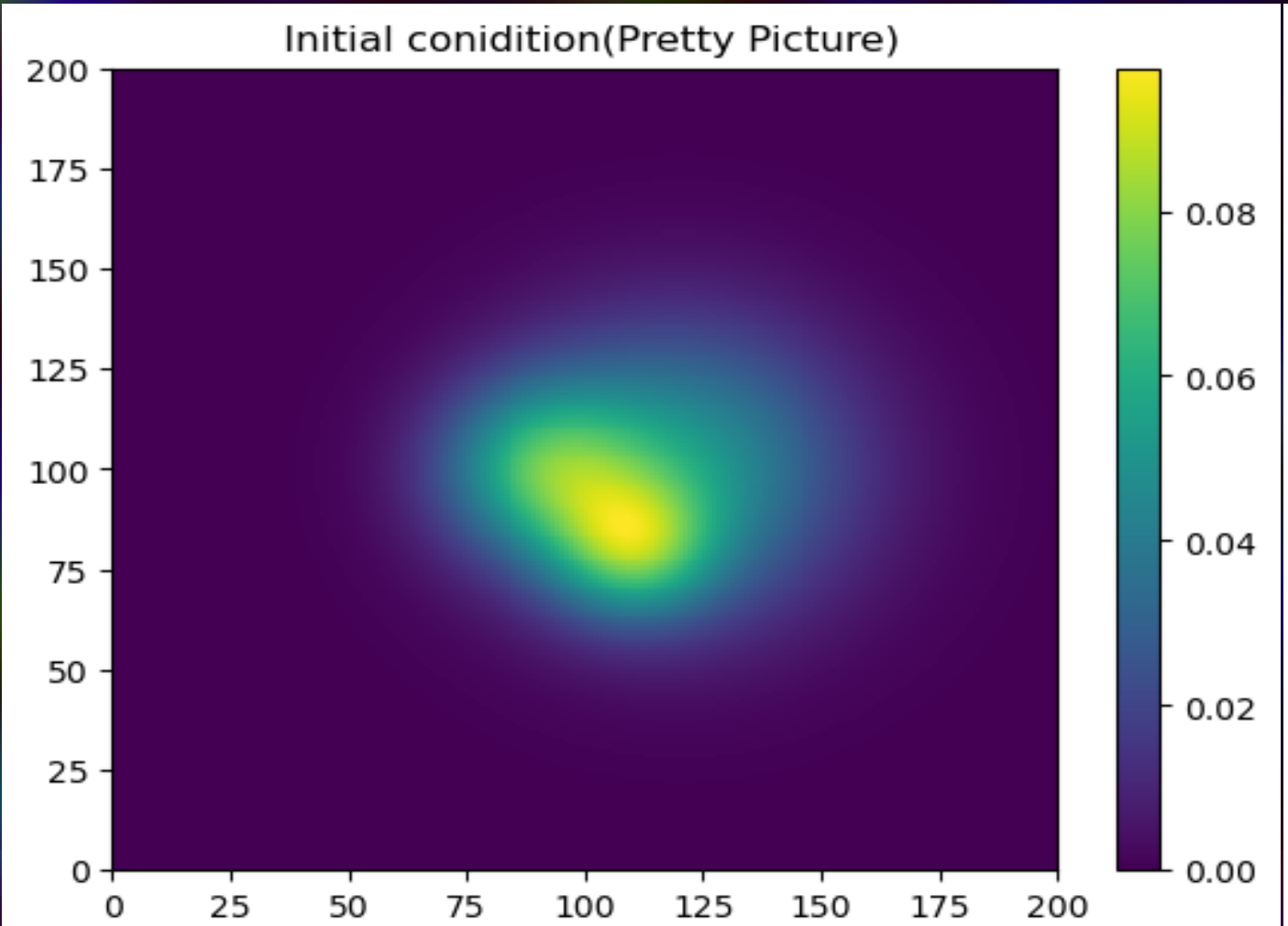
$$\partial_t v = \beta v n \left(1 - \frac{v}{V_{max}}\right)$$

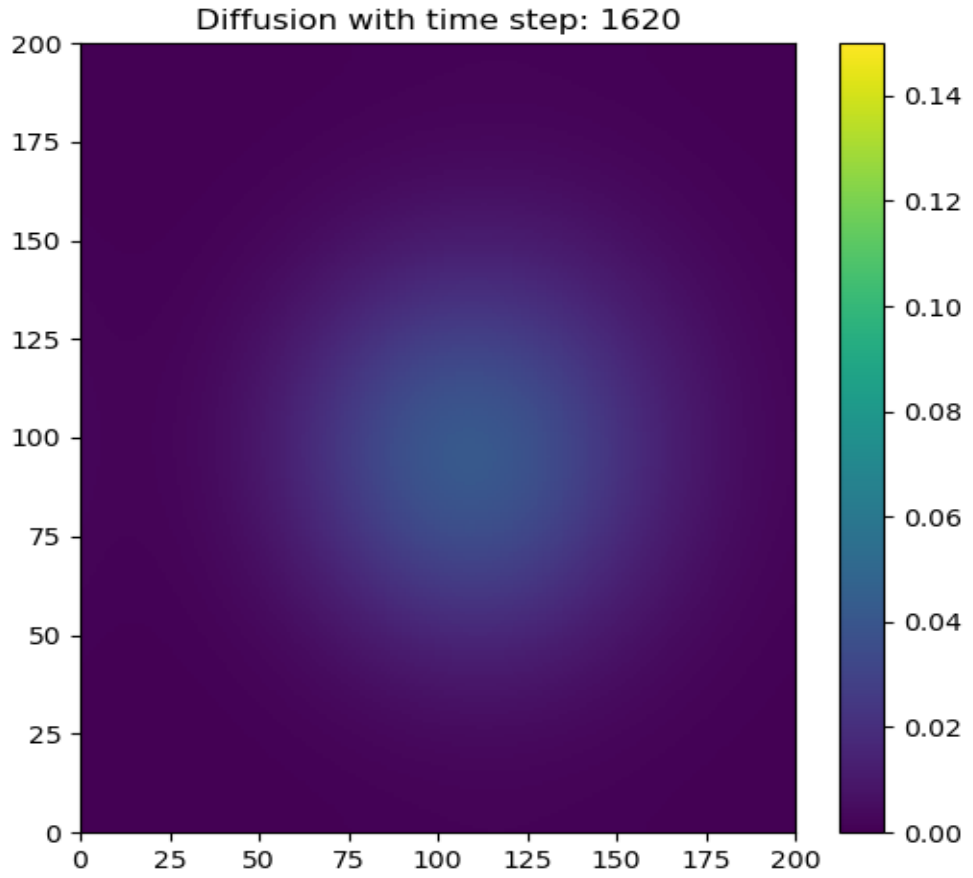
$$\partial_t n = D\nabla^2 n +$$

Diffusion

Only diffusion

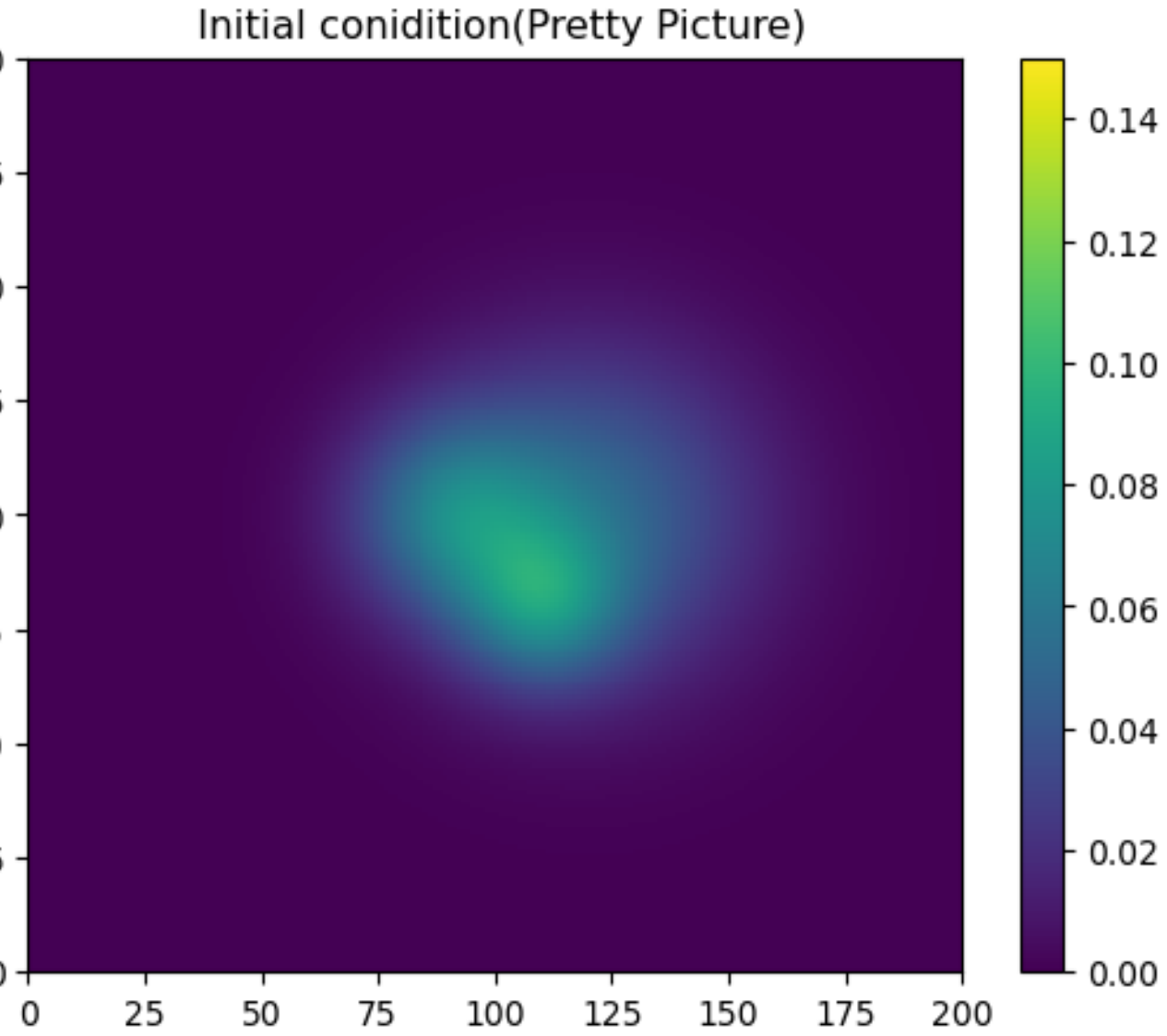
D constant = 2

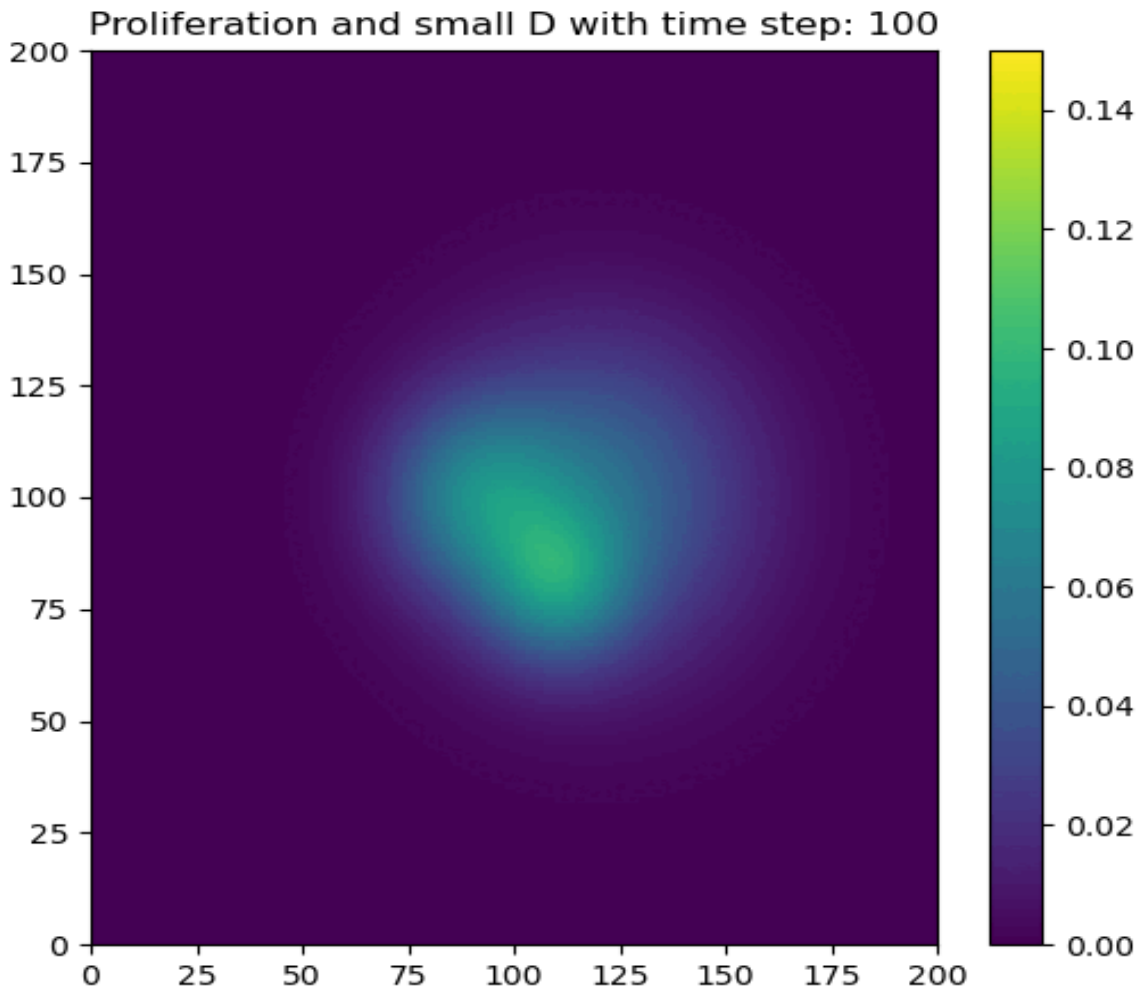




Proliferation and small diffusion

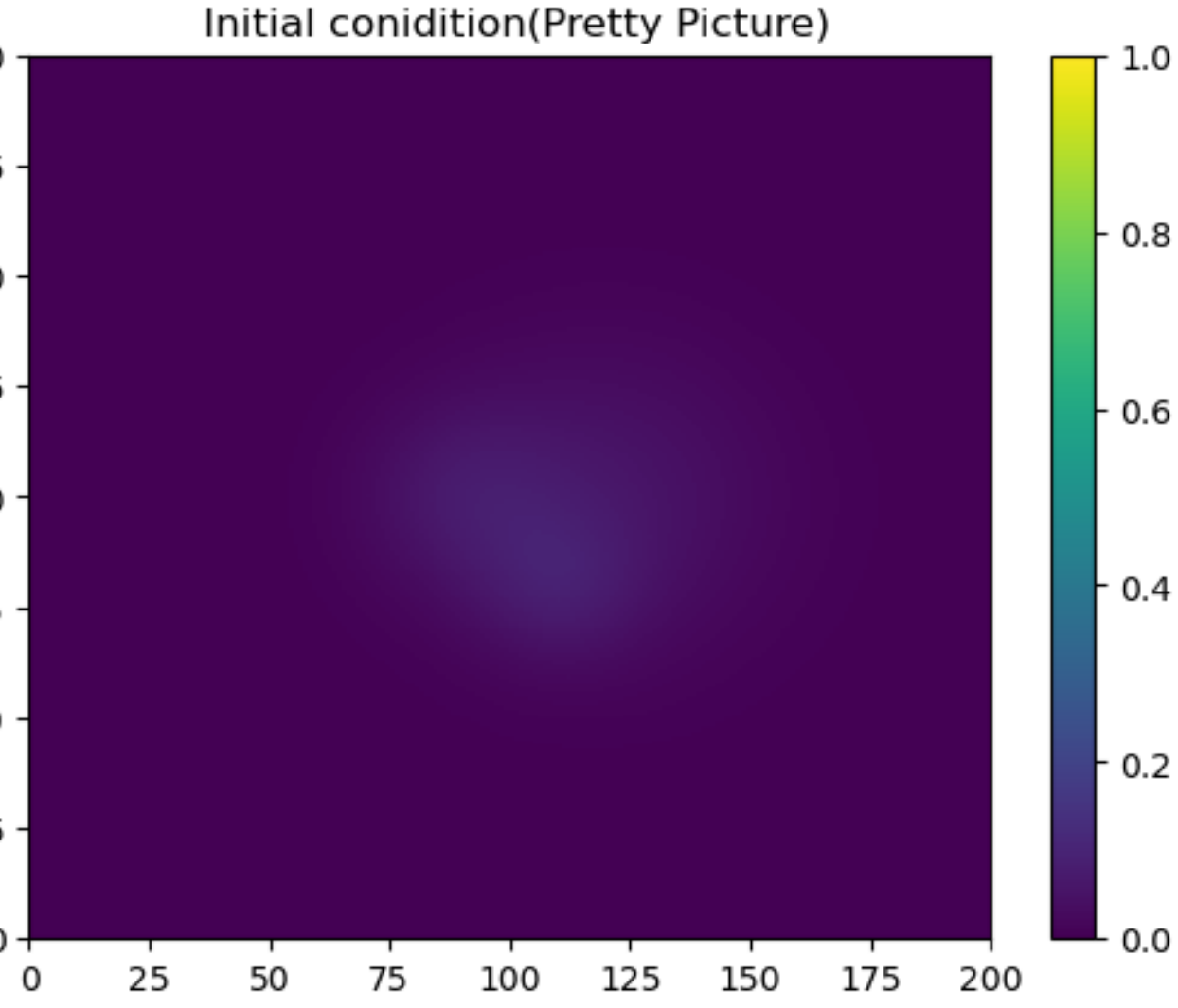
D constant = 0.001

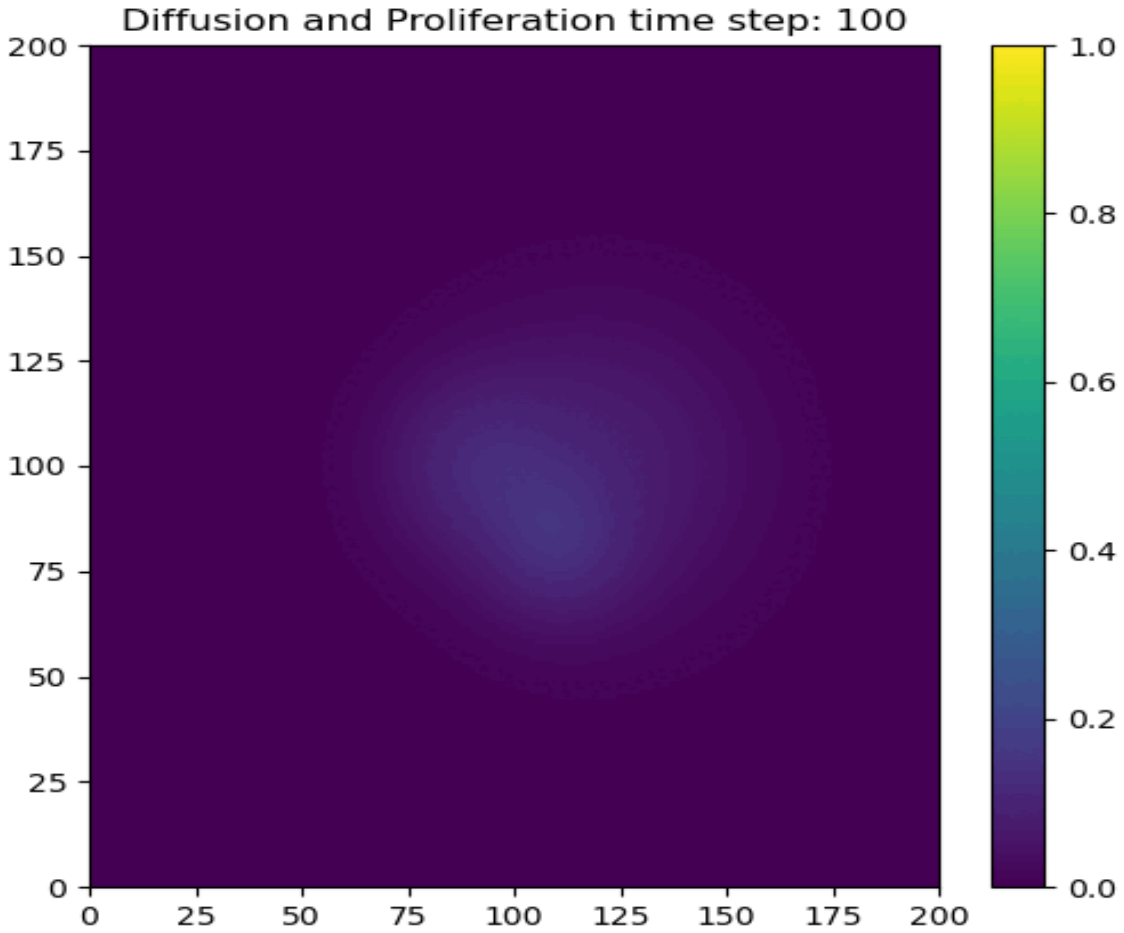




Proliferation and diffusion

D constant = 0.05





How does the vessel density affect the tumor growth?

Now Adding Vessels and evaluating how the tumor grows with small β (0.001) and larger β (0.1)

υ = Vessel density

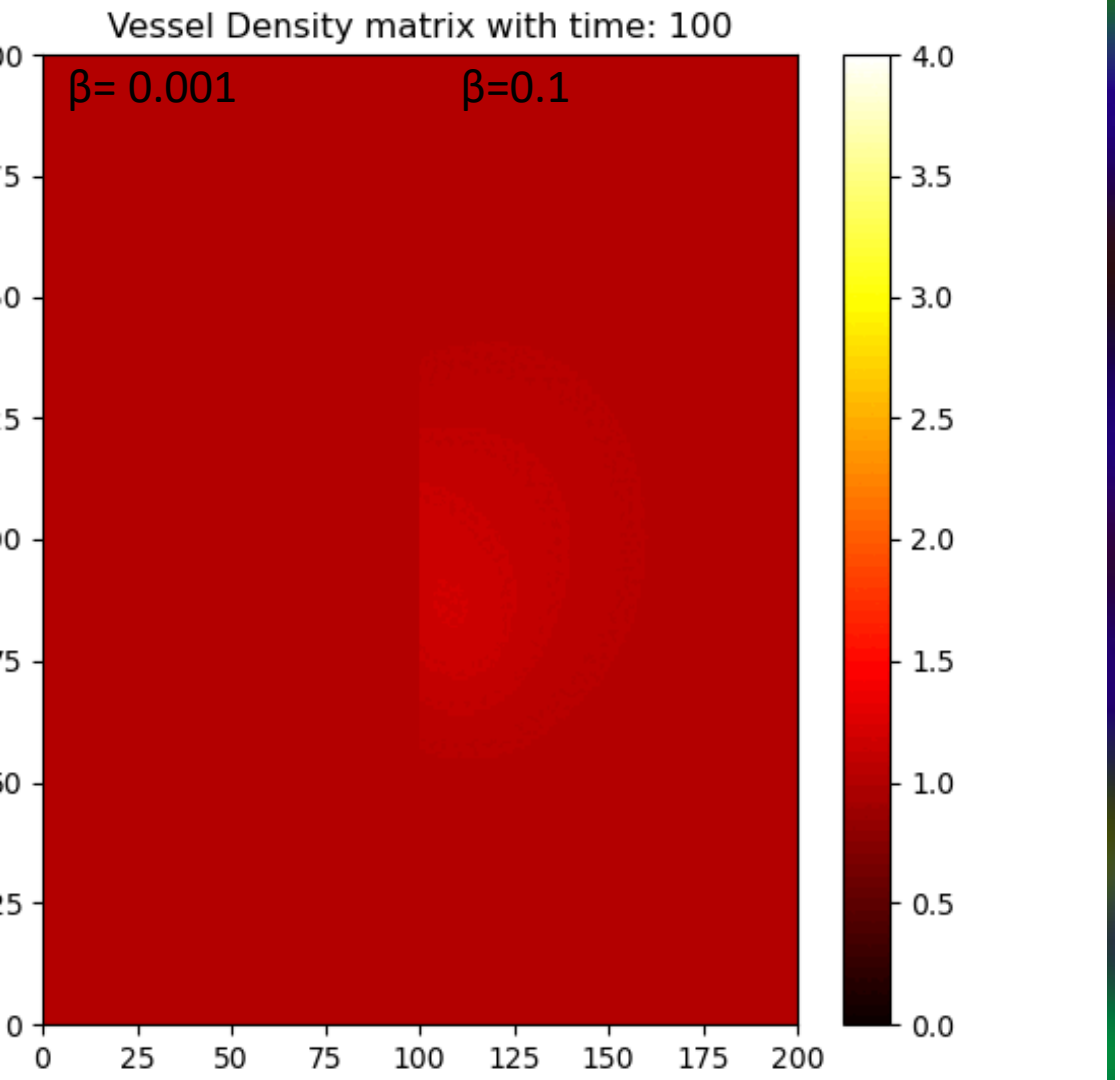
β = Vessel Growth

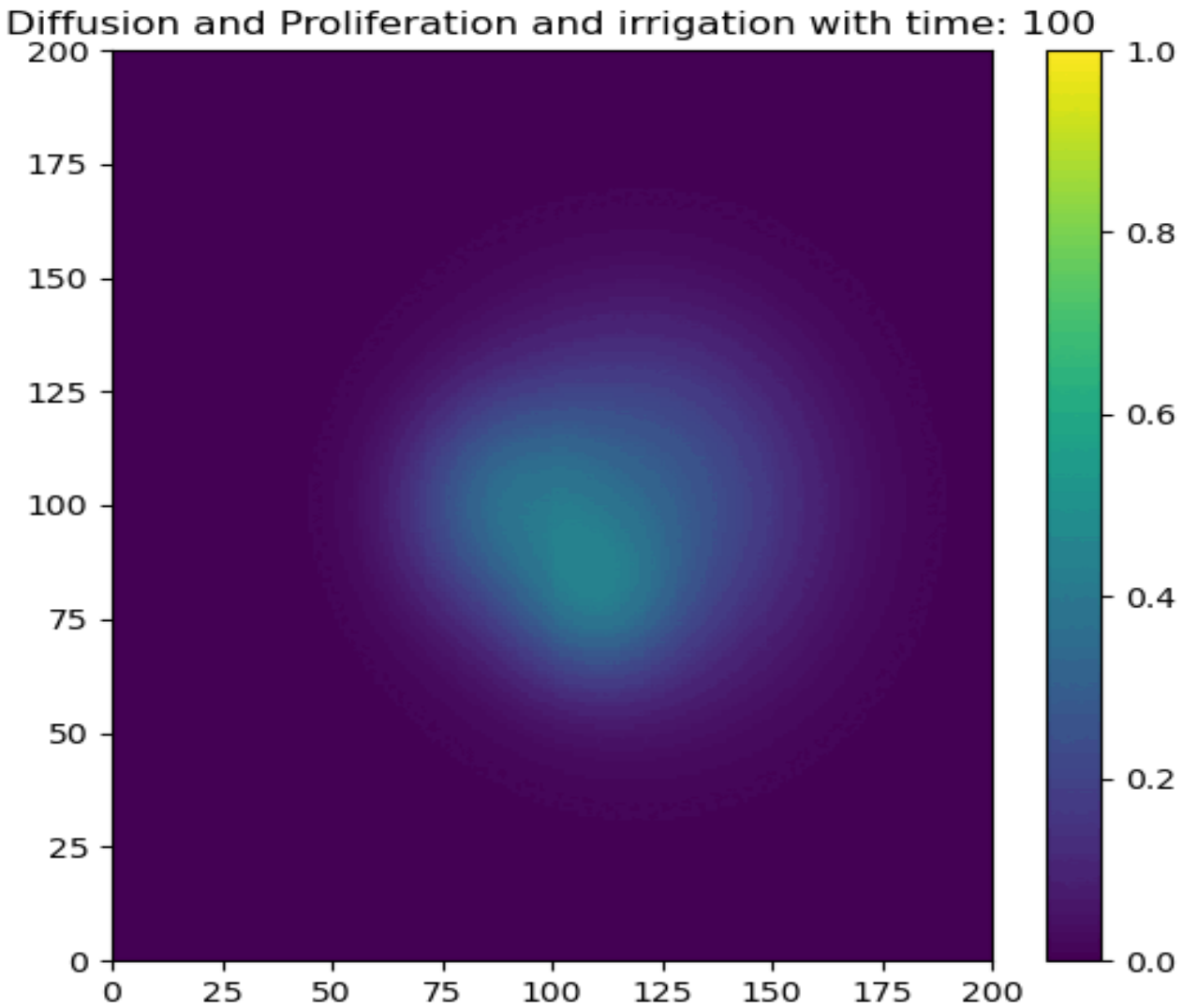
$$\partial_t v = \beta v n \left(1 - \frac{v}{V_{max}}\right)$$

it turns out more vessel density

More Tumor Growth

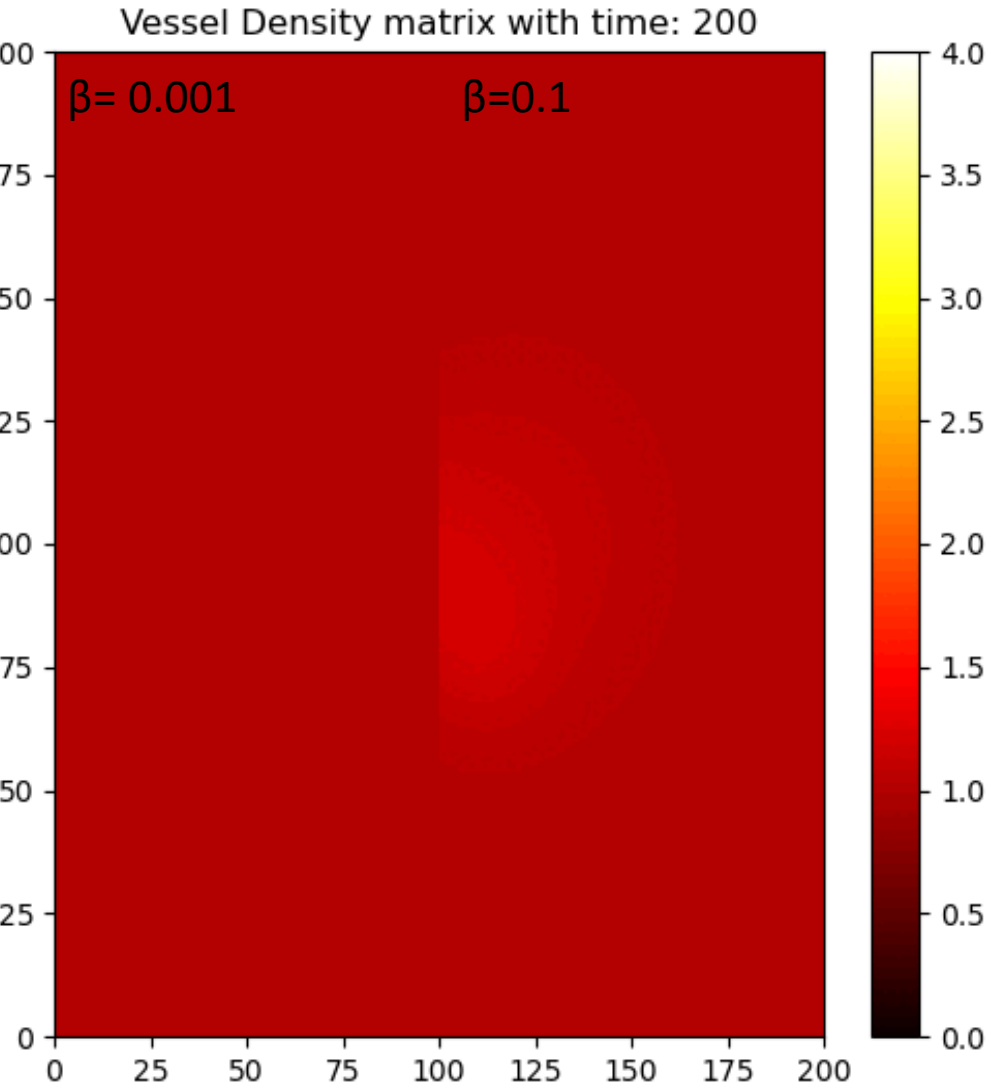
Diffusion constant = 0.2

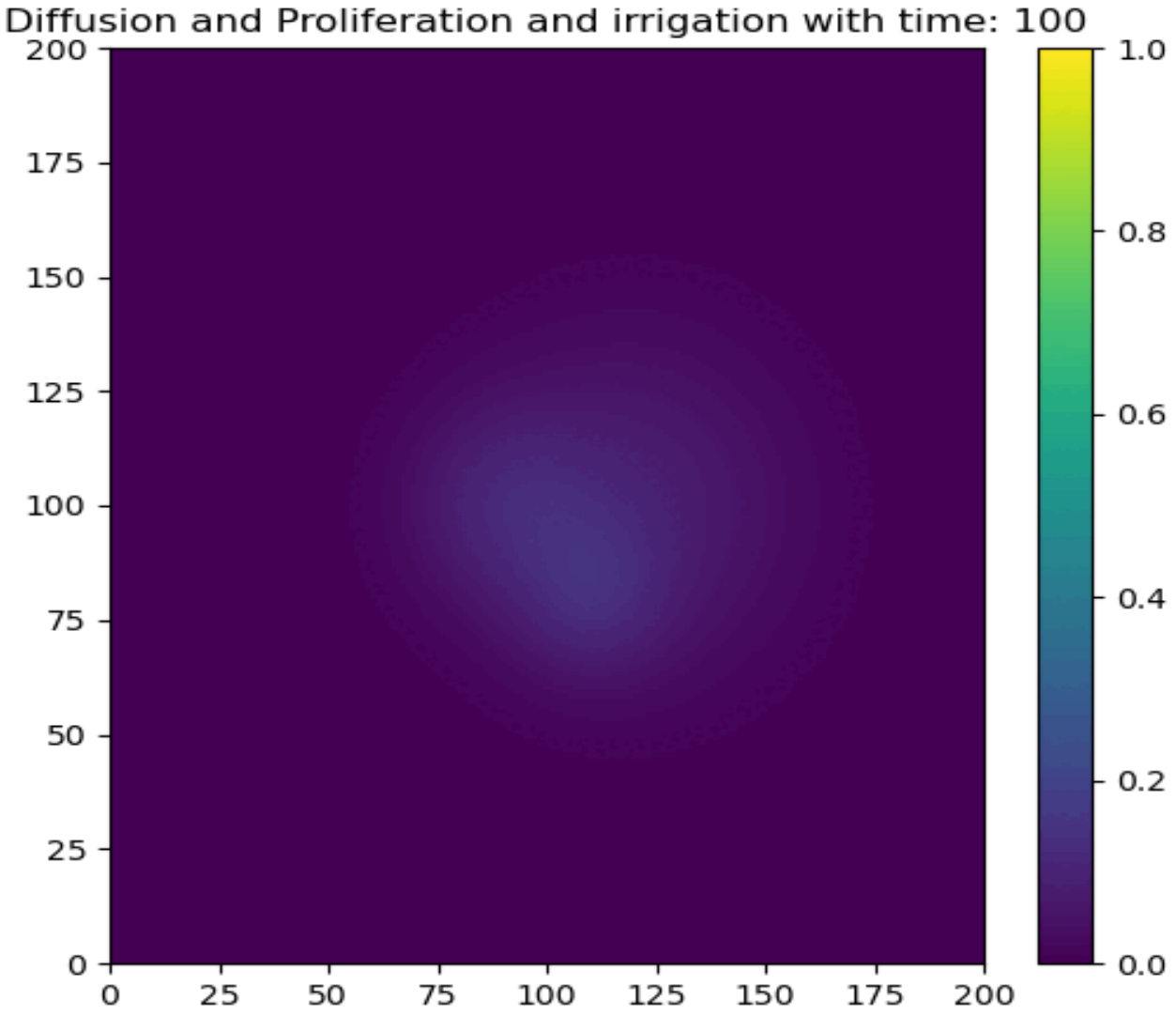




This time with smaller diffusion constant which shows

Diffusion constant = 0.05





Thank you <3

